Ramnath Kumar

◇ ramnathkumar181.github.io ◇ ramnathkumar181@gmail.com ◇ +14243714958 ◇ Google Scholar ◇

RESEARCH INTERESTS & SUMMARY

My research centers on Deep Learning, with a particular emphasis on developing **robust** and **efficient** methods for building practical, real-world systems. I am especially interested in *representation learning* and in rethinking the optimization strategies that underpin modern training paradigms - seeking approaches that are not only scalable and generalizable but also adaptive to the diverse demands of deployment environments.

EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, USA
PhD. in Computer Science — *09/2024 – Present*
. Advisor: Prof. Cho-Jui Hsieh

BITS Pilani, Hyderabad Campus — Hyderabad, India
B.E. in Computer Science, MSc. in Economics — *08/2016 – 08/2021*
. CGPA in Computer Science major: **9.65/10.0** (Top 10 in class of 255)
. Overall CGPA: 8.92/10.0

SELECTED RESEARCH/WORK EXPERIENCE

Google Research — Mountain View, USA
Student Researcher — *06/2026 – 09/2026*
. Mentors: Dr. Fei Wang and Prof. Inderjit S. Dhillon
. Identified catastrophic forgetting in RL post-training on vision-language models, where out-of-domain capabilities collapse while suite-wide averages mask degradation. Developed Content-Free Anchoring (CFA), a Fisher-weighted MMD penalty using only synthetic anchors that reduces worst-decile forgetting to $< 10\%$ with finite-sample guarantees, outperforming curated-data and other traditional baselines [1].

Google DeepMind — Bengaluru, India
Pre-Doctoral Researcher — *07/2022 – 08/2024*
. Mentors: Dr. Prateek Jain and Prof. Inderjit S. Dhillon
. Led a cross-functional team in developing and deploying RGD in product-driven research, improving performance of ViT by up to **1.01%** on ImageNet-1K [5].
. Developed an End-to-End efficient retrieval architecture (EHI), that improved upon prior retrieval benchmarks by up to **1.45%** at a fixed compute budget. This effort is slated for integration into various Google products [6].
. Earned recognition in Google's AdsML organization for advancing scalable product retrieval models and optimization strategies, improving relevance and efficiency across global ad-serving systems through involving Related Ads features in the query of the model.
. Took initiative in proposing a novel end-to-end retrieval architecture, improving efficiency and robustness in ongoing projects, with contributions recognized in . The approach has also been slated for integration with various teams including Keras, Youtube AutoML, and many more.

Google Research — Bengaluru, India
Research Associate — *04/2022 – 07/ 2022*
. Mentor: Dr. Dheeraj Nagaraj
. Developed Introspective Experience Replay (IER), a novel replay buffer sampling strategy inspired by Reverse Experience Replay (RER), demonstrating up to **7**× faster convergence in Reinforcement Learning algorithms such as DQN and TD3 [9].

Mila - Quebec Artificial Intelligence Institute Quebec, Canada
Consultant *07/2021 – 03/2022*
. Mentor: Prof. Yoshua Bengio
. Initiated and led a project exploring the impact of diversity in meta-learning, collaborating closely with his group. Designed and executed experiments, leading to an oral presentation at the "*Association for the Advancement of Artificial Intelligence*" (AAAI) that challenged the conventional wisdom of diversity being strictly useful to meta-learning [7].

Amazon ML Bengaluru, India
Applied Scientist Intern *01/2021 – 06/2021*
. Mentor: Dr. Gokul Swamy
. Led a study on causal attribution within Amazon's sales model as first author, with findings presented at Amazon's internal conference (AMLC 2021).

Kno.e.sis, Wright State University Dayton, USA
Research Intern *05/2019 – 08/2019*
. Mentors: Prof. Amit P. Sheth and Prof. Krishnaprasad Thirunarayan
. Collaborated with a cross-functional research team across multiple time zones as first author on the *eDarkFind* project, where I developed an unsupervised multi-view framework for Sybil detection, specifically targeting Darknet market activities [8].

PUBLICATIONS

* Indicates that the authors contributed equally to this work.

[1] CFA: Mitigating Forgetting in RL Post-Training with Content-Free Anchors
Ramnath Kumar, Jiong Zhang, Wei-Cheng Chang, Fei Wang, Hsiang-Fu Yu, Inderjit Dhillon, and Cho-Jui Hsieh
Under Review, ICLR 2027.

[2] FastLane: Efficient Routed Systems for Late-Interaction Retrieval
Ramnath Kumar, Prateek Jain, and Cho-Jui Hsieh
Under Review, EACL, 2027.

[3] MIRAGE: Protecting against Malicious Image Editing via False Moderation
Anshul Nasery, **Ramnath Kumar**, Cho-Jui Hsieh, and Sewoong Oh
Under Review, ICLR 2027.

[4] Flex-Act: Why Learn when you can Pick?
Ramnath Kumar, Kyle Ritscher, Junmin Zhu, Lawrence Liu, and Cho-Jui Hsieh
Transactions on Machine Learning Research, 2026.

[5] Stochastic Re-weighted Gradient Descent via Distributionally Robust Optimization
Ramnath Kumar, Kushal Alpesh Majmundar, Dheeraj Mysore Nagaraj, and Arun Suggala
Transactions on Machine Learning Research, 2024.
Google AI Blog Coverage.

[6] EHI: End-to-end Learning of Hierarchical Index for Efficient Dense Retrieval
Ramnath Kumar*, Anshul Mittal*, Nilesh Gupta, Aditya Kusupati, Inderjit Dhillon, and Prateek Jain
Transactions on Machine Learning Research, 2024.

[7] The Effect of diversity in Meta-Learning
Ramnath Kumar, Tristan Deleu, and Yoshua Bengio
Association for the Advancement of Artificial Intelligence (AAAI), 2023 (**Oral** Paper).
SyncedReview Blog Coverage.

[8] eDarkFind: Unsupervised Multi-view Learning for Sybil Account Detection
Ramnath Kumar, Shweta Yadav, Raminta Daniulaityte, Francois Lamy, Krishnaprasad Thirunarayan, Usha Lokala, and Amit Sheth
The Web Conference (WWW), 2020.

[9] Introspective Experience Replay: Look Back When Surprised
Ramnath Kumar and Dheeraj Nagaraj
Transactions on Machine Learning Research, 2024.
Google Research Blog Coverage.

[10] Rethinking Learning Dynamics in RL using Adversarial Networks
Ramnath Kumar, Tristan Deleu, and Yoshua Bengio
NeurIPS Workshop on DeepRL, 2022.

[11] Temporal Dynamics and Spatial Content in IoT Malware detection
Ramnath Kumar and G Geethakumari
IEEE TENCON, 2019.

SELECTED AWARDS AND HIGHLIGHTS

Quad Fellowship, Took part in a summit at the White House (U.S.A.) to discuss emerging technologies and their policy implications. 2024

Graduate Dean's Scholars Award, UCLA scholarship. 2024

Awardee, NTSE Scholar (India). 2014-2020